

Processando config 1, seed 4, teste: BlockAlphabit
xx
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
xx

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 2
Value of the statistic : 2.44
p-value of test : 0.29

CPU time used : 00:00:00.02

Generator state:

16777216 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 5.35
p-value of test : 0.72

CPU time used : 00:00:00.01

Generator state:

33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 124.44
p-value of test : 0.57

CPU time used : 00:00:00.02

Generator state:

50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32934.32
p-value of test : 0.26

CPU time used : 00:00:00.02

Generator state:

67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 8924.15
p-value of test : eps *****

CPU time used : 00:00:00.02

Generator state:

83886080 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom	:	305	
Chi-square statistic	:	3279.68	
p-value of test	:	eps	*****

CPU time used : 00:00:00.01

Generator state:

100663296 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic	:	0.42	
p-value of test	:	0.34	

CPU time used : 00:00:00.02

Generator state:

117440512 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32	
ChiSquare statistic	:	816.44	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	31	
ChiSquare statistic	:	1613.71	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	32	
ChiSquare statistic	:	433.89	
p-value of test	:	eps	*****

Test on the values of the Statistic R

Number of degrees of freedom	:	25	
ChiSquare statistic	:	72.14	
p-value of test	:	1.8e-6	*****

Test on the values of the Statistic C

Number of degrees of freedom	:	15	
ChiSquare statistic	:	45.82	
p-value of test	:	5.7e-5	*****

CPU time used : 00:00:00.07

Generator state:

134217728 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62	
ChiSquare statistic	:	564.47	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	62	
ChiSquare statistic	:	490.30	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	160	
ChiSquare statistic	:	197.18	
p-value of test	:	0.02	

Test on the values of the Statistic R

Number of degrees of freedom	:	57	
ChiSquare statistic	:	45.74	
p-value of test	:	0.86	

Test on the values of the Statistic C

Number of degrees of freedom	:	32	
ChiSquare statistic	:	34.18	
p-value of test	:	0.36	

CPU time used : 00:00:00.06

Generator state:

150994944 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.32
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):

(eps1 means a value < 1.0e-15):

	Test	p-value
5	HammingIndep, L = 16	eps
6	HammingIndep, L = 32	eps
8	RandomWalk1 H (L = 64)	eps
8	RandomWalk1 M (L = 64)	eps
8	RandomWalk1 J (L = 64)	eps
8	RandomWalk1 R (L = 64)	1.8e-6
8	RandomWalk1 C (L = 64)	5.7e-5
9	RandomWalk1 H (L = 320)	eps
9	RandomWalk1 M (L = 320)	eps

All other tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alhabit:  nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 2
```

smultin_MultinomialBitsOver test:

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1,  Mu = 2.9802323e-08,  Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom      : 2
Value of the statistic            : 2.10
p-value of test                   : 0.35
```

```
-----
CPU time used                     : 00:00:00.02
```

Generator state:

167772160 bits have been read.

```
*****
```

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 12.34
p-value of test : 0.14

CPU time used : 00:00:00.01

Generator state:

184549376 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128

Value of the statistic : 157.63
p-value of test : 0.04

CPU time used : 00:00:00.00

Generator state:
201326592 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32701.99
p-value of test : 0.60

CPU time used : 00:00:00.02

Generator state:
218103808 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers ≥ 10

```
-----  
Number of degrees of freedom      : 177  
Chi-square statistic              : 156.11  
p-value of test                   : 0.87  
-----
```

```
-----  
CPU time used                     : 00:00:00.01  
-----
```

Generator state:

234881024 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring_HammingIndep test:

```
-----  
N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0  
-----
```

Counters with expected numbers ≥ 10

```
-----  
Number of degrees of freedom      : 305  
Chi-square statistic              : 326.72  
p-value of test                   : 0.19  
-----
```

```
-----  
CPU time used                     : 00:00:00.01  
-----
```

Generator state:

251658240 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring_HammingCorr test:

```
-----  
N = 1, n = 524288, r = 0, s = 32, L = 32  
-----
```

```
-----  
Normal statistic                  : 0.68  
p-value of test                   : 0.25  
-----
```

CPU time used : 00:00:00.01

Generator state:
268435456 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	26.15
p-value of test	:	0.76

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	18.29
p-value of test	:	0.97

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	34.68
p-value of test	:	0.34

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	14.22
p-value of test	:	0.96

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	14.42
p-value of test	:	0.49

CPU time used : 00:00:00.07

Generator state:
285212672 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62	
ChiSquare statistic	:	768.28	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	62	
ChiSquare statistic	:	377.12	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	160	
ChiSquare statistic	:	265.16	
p-value of test	:	3.4e-7	*****

Test on the values of the Statistic R

Number of degrees of freedom	:	57	
ChiSquare statistic	:	56.79	
p-value of test	:	0.48	

Test on the values of the Statistic C

Number of degrees of freedom	:	32	
ChiSquare statistic	:	22.85	
p-value of test	:	0.88	

CPU time used : 00:00:00.05

Generator state:
301989888 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.26
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
9 RandomWalk1 H (L = 320)	eps
9 RandomWalk1 M (L = 320)	eps
9 RandomWalk1 J (L = 320)	3.4e-7

All other tests were passed

XXX
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
XXX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	2
Value of the statistic	:	0.088
p-value of test	:	0.96

CPU time used : 00:00:00.01

Generator state:
318767104 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 7.15
p-value of test : 0.52

CPU time used : 00:00:00.01

Generator state:
335544320 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256

Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 113.00
p-value of test : 0.83

CPU time used : 00:00:00.01

Generator state:
352321536 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32614.77
p-value of test : 0.72

CPU time used : 00:00:00.03

Generator state:
369098752 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 201.06
p-value of test : 0.10

CPU time used : 00:00:00.00

Generator state:

385875968 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 6380.61
p-value of test : eps *****

CPU time used : 00:00:00.01

Generator state:

402653184 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 0.023
p-value of test : 0.49

CPU time used : 00:00:00.00

Generator state:

419430400 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom : 32
ChiSquare statistic : 2737.79
p-value of test : eps *****

Test on the values of the Statistic M

Number of degrees of freedom : 31
ChiSquare statistic : 775.85
p-value of test : eps *****

Test on the values of the Statistic J

Number of degrees of freedom : 32
ChiSquare statistic : 144.72
p-value of test : 3.3e-16 *****

Test on the values of the Statistic R

Number of degrees of freedom : 25
ChiSquare statistic : 28.65
p-value of test : 0.28

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	21.41
p-value of test	:	0.12

CPU time used : 00:00:00.07

Generator state:
436207616 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	923.23
p-value of test	:	eps *****

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	396.48
p-value of test	:	eps *****

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	336.85
p-value of test	:	1.1e-14 *****

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	51.91
p-value of test	:	0.67

Test on the values of the Statistic C

Number of degrees of freedom : 32
ChiSquare statistic : 23.69
p-value of test : 0.86

CPU time used : 00:00:00.06

Generator state:
452984576 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.26
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

	Test	p-value
6	HammingIndep, L = 32	eps
8	RandomWalk1 H (L = 64)	eps
8	RandomWalk1 M (L = 64)	eps
8	RandomWalk1 J (L = 64)	3.3e-16
9	RandomWalk1 H (L = 320)	eps
9	RandomWalk1 M (L = 320)	eps
9	RandomWalk1 J (L = 320)	1.1e-14

All other tests were passed

XX

Starting Alphabit: nb = 16777216

Version: TestU01 1.2.3

XX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216

Number of cells = $2^L =$ 4
Expected number per cell = 4194304
Hashing = FALSE

For $\Delta > -1$, we use the ChiSquare approximation
Correction factor of the ChiSquare:

$\Delta = 1$, $\mu = 2.9802323e-08$, $\sigma = 1$

Test Results for $\Delta = 1.0000$

Number of degrees of freedom	:	2
Value of the statistic	:	1.22
p-value of test	:	0.54

CPU time used : 00:00:00.01

Generator state:
469761792 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = $2^L =$ 16
Expected number per cell = 1048576
Hashing = FALSE

For $\Delta > -1$, we use the ChiSquare approximation
Correction factor of the ChiSquare:

$\Delta = 1$, $\mu = 2.0861626e-07$, $\sigma = 1$

Test Results for $\Delta = 1.0000$

Number of degrees of freedom	:	8
Value of the statistic	:	11.37
p-value of test	:	0.18

CPU time used : 00:00:00.00

Generator state:
486539008 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 119.70
p-value of test : 0.69

CPU time used : 00:00:00.02

Generator state:

503316224 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

```
Number of degrees of freedom      : 32768
Value of the statistic            : 32539.58
p-value of test                   : 0.81
```

```
-----
CPU time used                     : 00:00:00.01
```

```
Generator state:
520093440 bits have been read.
```

```
*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8
```

```
sstring_HammingIndep test:
```

```
-----
N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0
```

```
Counters with expected numbers >= 10
```

```
-----
Number of degrees of freedom      : 177
Chi-square statistic              : 12877.05
p-value of test                   : eps      *****
```

```
-----
CPU time used                     : 00:00:00.01
```

```
Generator state:
536870656 bits have been read.
```

```
*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8
```

```
sstring_HammingIndep test:
```

```
-----
N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0
```

```
Counters with expected numbers >= 10
```

```
-----
Number of degrees of freedom      : 305
Chi-square statistic              : 5159.60
p-value of test                   : eps      *****
-----
```

CPU time used : 00:00:00.01

Generator state:

553647872 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic	:	2.84
------------------	---	------

p-value of test	:	2.2e-3
-----------------	---	--------

CPU time used	:	00:00:00.00
---------------	---	-------------

Generator state:

570425088 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
------------------------------	---	----

ChiSquare statistic	:	2658.61
---------------------	---	---------

p-value of test	:	eps	*****
-----------------	---	-----	-------

Test on the values of the Statistic M

Number of degrees of freedom	:	31
------------------------------	---	----

ChiSquare statistic	:	1381.90
---------------------	---	---------

p-value of test	:	eps	*****
-----------------	---	-----	-------

Test on the values of the Statistic J

Number of degrees of freedom	:	32	
ChiSquare statistic	:	408.49	
p-value of test	:	eps	*****

Test on the values of the Statistic R

Number of degrees of freedom	:	25	
ChiSquare statistic	:	37.24	
p-value of test	:	0.05	

Test on the values of the Statistic C

Number of degrees of freedom	:	15	
ChiSquare statistic	:	11.74	
p-value of test	:	0.70	

CPU time used : 00:00:00.07

Generator state:
587202304 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62	
ChiSquare statistic	:	1238.35	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	62	
ChiSquare statistic	:	616.61	
p-value of test	:	eps	*****

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 2
Value of the statistic : 0.94
p-value of test : 0.62

CPU time used : 00:00:00.01

Generator state:

620756480 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 10.89
p-value of test : 0.21

CPU time used : 00:00:00.00

Generator state:
637533696 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 135.12
p-value of test : 0.32

CPU time used : 00:00:00.01

Generator state:
654310912 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

```

-----
N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:
Delta = 1, Mu = 0.00097653271, Sigma = 1

```

```

-----
Test Results for Delta = 1.0000

```

```

Number of degrees of freedom : 32768
Value of the statistic : 32955.05
p-value of test : 0.23

```

```

-----
CPU time used : 00:00:00.02

```

```

Generator state:
671088128 bits have been read.

```

```

*****
HOST = matheus-IdeaPad-3-15ALC6, Linux

```

```

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

```

```

sstring_HammingIndep test:

```

```

-----
N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

```

```

Counters with expected numbers >= 10

```

```

-----
Number of degrees of freedom : 177
Chi-square statistic : 142.21
p-value of test : 0.97

```

```

-----
CPU time used : 00:00:00.01

```

```

Generator state:
687865344 bits have been read.

```

```

*****
HOST = matheus-IdeaPad-3-15ALC6, Linux

```

```

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

```

unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 307.58
p-value of test : 0.45

CPU time used : 00:00:00.01

Generator state:

704642560 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 0.96
p-value of test : 0.17

CPU time used : 00:00:00.00

Generator state:

721419776 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	39.45
p-value of test	:	0.17

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	38.89
p-value of test	:	0.16

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	48.79
p-value of test	:	0.03

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	36.80
p-value of test	:	0.06

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	25.94
p-value of test	:	0.04

CPU time used : 00:00:00.07

Generator state:
738196992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom : 62
ChiSquare statistic : 1871.80
p-value of test : eps *****

Test on the values of the Statistic M

Number of degrees of freedom : 62
ChiSquare statistic : 898.91
p-value of test : eps *****

Test on the values of the Statistic J

Number of degrees of freedom : 160
ChiSquare statistic : 288.46
p-value of test : 2.1e-9 *****

Test on the values of the Statistic R

Number of degrees of freedom : 57
ChiSquare statistic : 61.52
p-value of test : 0.32

Test on the values of the Statistic C

Number of degrees of freedom : 32
ChiSquare statistic : 49.60
p-value of test : 0.02

CPU time used : 00:00:00.05

Generator state:
754973952 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.25
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
9 RandomWalk1 H (L = 320)	eps

```
9 RandomWalk1 M (L = 320)          eps
9 RandomWalk1 J (L = 320)          2.1e-9
-----
```

All other tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alhabit:  nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
```

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen:  r = 0,    s = 32,    w = 32
```

smultin_MultinomialBitsOver test:

```
-----
```

```
  N =  1,  n = 16777216,  r =  0,    s = 32,    L =    2,
    Sparse = FALSE
```

```
    Number of bits = n = 16777216
    Number of cells = 2^L =                      4
    Expected number per cell =      4194304
    Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
    Delta =    1,    Mu = 2.9802323e-08,    Sigma =          1
```

```
-----
```

Test Results for Delta = 1.0000

```
Number of degrees of freedom      :    2
Value of the statistic             :    2.16
p-value of test                   :    0.34
```

```
-----
```

CPU time used : 00:00:00.02

Generator state:

771751168 bits have been read.

```
*****
```

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen:  r = 0,    s = 32,    w = 32
```

smultin_MultinomialBitsOver test:

```
-----
```

```
  N =  1,  n = 16777216,  r =  0,    s = 32,    L =    4,
    Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 2.0861626e-07, Sigma = 1
```

Test Results for Delta = 1.0000

```
Number of degrees of freedom : 8
Value of the statistic       : 7.60
p-value of test              : 0.47
```

CPU time used : 00:00:00.00

Generator state:

788528384 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32
```

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

```
Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 3.784895e-06, Sigma = 1
```

Test Results for Delta = 1.0000

```
Number of degrees of freedom : 128
Value of the statistic       : 127.15
p-value of test              : 0.50
```

CPU time used : 00:00:00.00

Generator state:

805305600 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32447.36
p-value of test : 0.90

CPU time used : 00:00:00.02

Generator state:

822082816 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 139.03
p-value of test : 0.98

CPU time used : 00:00:00.01

Generator state:
838860032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom	:	305
Chi-square statistic	:	303.98
p-value of test	:	0.51

CPU time used : 00:00:00.00

Generator state:
855637248 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 1.29
p-value of test : 0.10

CPU time used : 00:00:00.01

Generator state:
872414464 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32
```

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	57.23
p-value of test	:	4.0e-3

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	18.42
p-value of test	:	0.96

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	44.36
p-value of test	:	0.07

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	33.64
p-value of test	:	0.12

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	8.88
p-value of test	:	0.88

CPU time used : 00:00:00.07

Generator state:

889191680 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32
```

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62	
ChiSquare statistic	:	2007.12	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	62	
ChiSquare statistic	:	991.06	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	160	
ChiSquare statistic	:	288.28	
p-value of test	:	2.1e-9	*****

Test on the values of the Statistic R

Number of degrees of freedom	:	57	
ChiSquare statistic	:	42.84	
p-value of test	:	0.92	

Test on the values of the Statistic C

Number of degrees of freedom	:	32	
ChiSquare statistic	:	46.78	
p-value of test	:	0.04	

CPU time used : 00:00:00.05

Generator state:
905968640 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3

Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.25
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
-----	-----
9 RandomWalk1 H (L = 320)	eps
9 RandomWalk1 M (L = 320)	eps
9 RandomWalk1 J (L = 320)	2.1e-9
-----	-----

All other tests were passed